
Beyond ARC and GSM: Langford Coverage as a Benchmark for ASI

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ABSTRACT

As foundational models approach Artificial Superintelligence (ASI), standard intelligence evaluations like the Massive Multitask Language Understanding (MMLU), Grade School Math 8K (GSM8K), and the Abstraction and Reasoning Corpus (ARC-AGI) fail to adequately differentiate between true algorithmic reasoning and sophisticated statistical correlation. We introduce "Langford Coverage," an infinite, procedurally generated evaluation suite based on the Toroidal Residue-Covering Benchmark (TRCB). By mapping arithmetic intervals to multi-dimensional punctured toroidal grids via the Chinese Remainder Theorem, this benchmark scales the canonical search space beyond physical limits, surpassing 10^{119} states by the 17th prime modulus. Furthermore, the benchmark utilizes a Dynamic Entry Protocol, forcing agents to balance computational risk against reward. Langford Coverage provides a mathematically rigorous, physically un-cheatable protocol to verify whether an AI model possesses the fluid intelligence and theorem-discovery capabilities requisite for ASI.

1. The End of Empirical Benchmarking

The rapid advancement of Large Language Models (LLMs) has saturated existing benchmarks, with models frequently achieving scores above 90% on exams like MMLU and GSM8K. However, recent empirical evaluations, such as Apple's study on the GSM-Symbolic benchmark, demonstrate that modifying non-logical variables in these word problems causes performance to plummet by up to 65%. Current models largely rely on rote memorization and associative pattern-matching from pre-training corpora, rather than executing "System 2" cognitive reasoning and executing algorithmic logic from first principles.

While visual benchmarks like ARC-AGI attempt to evaluate fluid intelligence via grid transformations, their underlying rules are mathematically informal and rely heavily on human cognitive priors. To accurately verify the emergence of ASI, the field requires a framework governed by a closed-form combinatorial theory that physically prevents data contamination.

2. The Mathematical Foundation: Residue-Covering Isomorphisms

Langford Coverage is built upon the complete combinatorial classification of the finite residue-covering problem on the interval $I = [34, 66] \cap \mathbb{Z}$. While exhaustive computer searches over the 10,080 ordered configurations easily reveal that a maximum of 32 out of 33 integers can be covered under strict prime multiplicity constraints, computation alone fails to illuminate the underlying structural mechanics.

The mathematical reduction bypasses this computational brute-force approach entirely:

- ▶ **Toroidal Mapping:** The interval is mapped onto a 5×7 grid utilizing the Chinese Remainder Theorem, establishing a perfect spatial bijection. This mapping contains exactly two punctures, located at coordinates (2, 4) and (3, 5), which correspond to the missing integer residue classes.
- ▶ **Seed Deletions:** Modulo 5 and 7 selections act as a "seed" that geometrically deletes rows and columns. This elimination creates a residual board $R(B)$ of surviving active cells.
- ▶ **The Singleton-Class Theorem:** Modulo 2 and 3 completions are isomorphic to selecting a mod-6 coloring scheme on the grid. An assignment achieves optimal coverage if and only if the chosen omitted color is a singleton—appearing exactly once—on the residual board.

By analytically deriving the seed distribution and isolating the 126 canonical optimal assignments using exact grid geometry, this proof demonstrates how abstract numerical constraints can be perfectly translated into spatial logic.

3. Chrono-Scaling and the Cosmic Memorization Barrier

To transform this mathematical proof into an ASI evaluation framework, the prime moduli dimensions are dynamically scaled. For each prime p_i , a multiplicity constraint m_i dictates the number of distinct residue classes an agent is allowed to select.

By setting the multiplicity constraint to the middle binomial coefficient, the number of valid configuration states experiences an exponential combinatorial explosion.

At $k = 15$ primes, the canonical search space hits 9.507×10^{87} configurations. This scale requires 10^7 times more bytes to store than the total estimated number of atoms in the observable universe.

At $k = 17$ primes, the configuration states explode to 5.472×10^{119} .

This aggressive scaling introduces the "Cosmic Memorization Barrier". Due to the Bekenstein bound, storing a pre-computed lookup table of these solutions is a physical impossibility. Consequently, an ASI candidate cannot succeed via data contamination; it must autonomously discover and execute the underlying structural mathematics.

4. Langford Coverage Evaluation Protocol for ASI

The benchmark operates using a game-theoretic "Dynamic Entry Protocol". The ASI candidate must self-select a starting difficulty grid size and balance its computational costs against exponential time penalties.

Candidate ASI models are profiled across three tiers:

TIER 1 — ISOMORPHIC TRANSLATION

The model must correctly map a scaled interval to a toroidal grid and calculate the exact coordinates of the punctures.

TIER 2 — CONSTRAINT OPTIMIZATION

Given a specific seed of deleted rows and columns, the model must construct the residual board and identify all singleton colors. This directly evaluates the model's capacity to execute complex search and pruning algorithms without getting lost in combinatorial depth.

TIER 3 — AUTONOMOUS THEOREM DISCOVERY

The ASI candidate is provided with a training set of smaller, completed residue-covering puzzles. It must inductively discover formal mathematical rules, such as the Singleton-Class Theorem, and write optimized programming logic to solve massive grids instantly.

5. Conclusion

The Langford Coverage framework bridges the gap between raw algorithmic calculation and abstract spatial reasoning. By demanding true isomorphic fluidity under the immutable constraints of physical limits and computational complexity, this benchmark provides a procedurally generated, un-cheatable metric to verify the cognitive emergence of Artificial Superintelligence.